# Video 17: *Implicit Finite Difference Equations for Parabolic Equations*

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# Unsteady Heat Conduction equation

We will re-examine the unsteady heat conduction equation (parabolic equation):

$$\frac{\partial u}{\partial t} - \kappa \nabla^2 u = f$$

Previously we looked at a "forward difference":

$$\frac{\partial u}{\partial t} \rightarrow \frac{u_{t+\Delta t,x} - u_{t,x}}{\Delta t}$$

This unit, we will consider what would happen if we selected a "backward" difference:

$$\frac{\partial u}{\partial t} \rightarrow \frac{u_{t,x} - u_{t-\Delta t,x}}{\Delta t}$$

# Unsteady Heat Conduction Equation: Time discretization

- Let's start with the time discretization:
- Let's divide time into $\Delta t$ intervals represented by the $k$-index
- Using a backward difference in time:

$$\frac{\partial u_i^k}{\partial t} \rightarrow \frac{u_i^k - u_i^{k-1}}{\Delta t}$$

- Again, with subscript indexes:

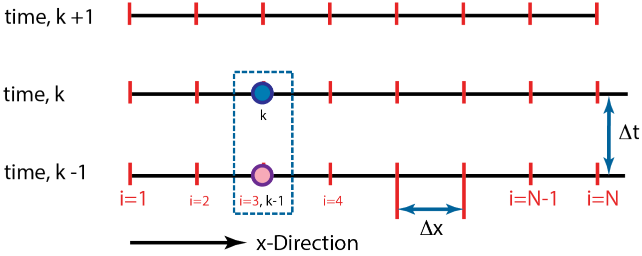
$$\frac{\partial u_{i,k}}{\partial t} \rightarrow \frac{u_{i,k} - u_{i,k-1}}{\Delta t}$$

# Time discretization: Graphical

- So, the equation:

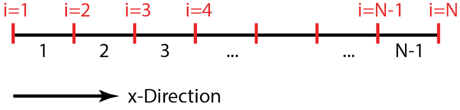
$$\frac{\partial u_{i,k}}{\partial t} \rightarrow \frac{u_{i,k} - u_{i,k-1}}{\Delta t}$$

- As an image (notice that time $k$ is the middle row):



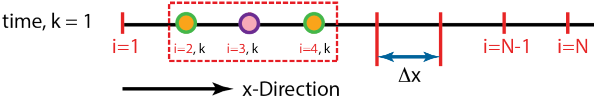
# Unsteady Heat Conduction Equation: Discretize the Domain

- We once again divide the domain into $N-1$ intervals of length $\Delta x$, with $N$ nodes.




- We know how to discretize the 1-D spatial derivative, so let's start there – using the index-$i$:

$$\nabla^2 u_{k,i} = \frac{\partial^2 u_{k,i}}{\partial x^2} \to \frac{u_{k,i-1} - 2u_{k,i} + u_{k,i+1}}{\Delta x^2}$$

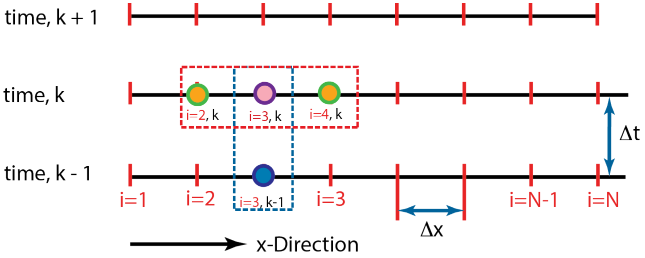


# Overall Finite Difference Equation for a Node

- Discrete equation compiled:

$$\left(\frac{u_{i,k} - u_{i,k-1}}{\Delta t}\right) - \kappa\left(\frac{u_{i-1,k} - 2u_{i,k} + u_{i+1,k}}{\Delta x^2}\right) = f_{i,k}$$

- As an image:

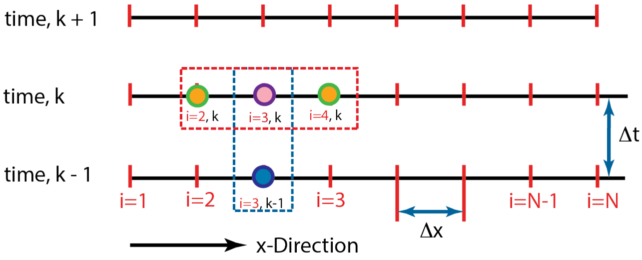


# Overall Finite Difference Equation for a Node

- Re-arrange the equation

$$u_{i,k} = u_{i,k-1} + \frac{\kappa \Delta t}{\Delta x^2}\left(u_{i-1,k} - 2u_{i,k} + u_{i+1,k}\right) + f_{i,k}\Delta t$$

- As an image:



# Nodal update equation

- The problem is, we are looking for $u_{i,k}$ using $u_{i,k-1}$ and $\frac{\partial^2 u}{\partial x^2}$ at $t = k$
- We know $u_{i,k-1}$, but we don't know $\frac{\partial^2 u}{\partial x^2}$ at $t = k$ until we know $u_{i,k}$
- This means we need to solve simultaneously for $u_{i,k}$ and $\frac{\partial^2 u}{\partial x^2}$.
- Ie. it is probably better to write our unknowns on the left and knowns on the right side of the equation:

$$u_{i,k} - \frac{\kappa \Delta t}{\Delta x^2}\left(u_{i-1,k} - 2u_{i,k} + u_{i+1,k}\right) = u_{i,k-1} + f_{i,k}\Delta t$$

- Where, the LHS is the linear system of equations for the unknown and RHS is the known values ($u_{i,k-1}$ and $f$)

# Nodal update equation

- This equation:

$$u_{i,k} - \frac{\kappa \Delta t}{\Delta x^2}\left(u_{i-1,k} - 2u_{i,k} + u_{i+1,k}\right) = u_{i,k-1} + f_{i,k}\Delta t$$

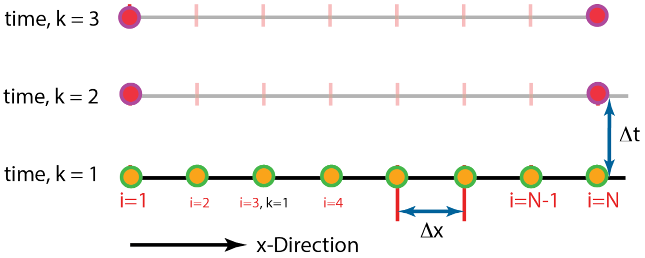
- Looks like the following equations in matrix form:

$$\left[\mathbb{I} - \left(\frac{\kappa \Delta t}{\Delta x^2}\right) A_{[1,-2,1]}\right][u_k] = [u_{k-1} + f\Delta t] \quad (1)$$

- Where, $\mathbb{I}$ is the identity matrix.
- **See if you can convince yourself that of the matrix structure shown before next class!!**

# Setup initial conditions and boundary conditions

- The whole solution still relies on the fact that you know the solution/temperature at some initial time $\rightarrow$ initial condition (I.C.)
- The ends/boundaries of the bar also need *fixed values* or boundary conditions.



# Psuedo Code and Matlab Example

```
1 - Setup your spatial discretization
 --- deltaX = 1/(N-1)

2 - Setup your temporal discretization
  --- deltat = dt; tEnd = Value;


3 - Setup the initial conditions in the bar
 ---   u(1, :) = Initial Condition in the bar

4 - Loop over time
  --- While (k > tEnd)
  --- Setup an A matrix
  --- Setup an Identity matrix
  --- Setup the boundary conditions

4.1 - Set the boundary conditions

--- A(1,:) = 0
--- A(1,1) = 1
--- A(end,:) = 0
--- A(end,end) = 1
--- RHS(1,k) = Value\_LHS
--- RHS(end,k) = Value\_RHS


4.2 - Solve for $u_{k}$ by solving Au = RHS


4.3 Loop back through time for the next time step.

Done!
```

# What have we learned

- The process for an implicit parabolic equation solution is VERY similar to elliptic equations
- Need to march in one direction with the equation (to advance in time or one direction in space)
- We need to solve a linear system at each time-step – for the implicit approach.
- This is called an implicit method – unconditionally stable, harder to setup, but beneficial in the end!